Performance Testing Report

Project Title

Strategic Product Placement Analysis using Tableau

1. Introduction

Performance testing was conducted to evaluate the responsiveness, usability, and efficiency of the Tableau dashboards developed for the Strategic Product Placement Analysis project. The goal of this testing was to ensure that all visualizations, filters, and interactions work smoothly without delays and provide a seamless analytical experience for users.

2. Objectives of Performance Testing

- Verify dashboard loading performance
- Test responsiveness of interactive filters
- Validate accuracy of visualizations and calculated fields
- Ensure smooth user interaction with charts and dashboards
- Evaluate overall usability of the Tableau dashboards

3. Test Environment

Tool Used: Tableau Public

Dataset: Retail Product Sales Dataset

Operating System: Windows 11

Processor: 13th Gen Intel(R) Core(TM) i5-13500H (2.60 GHz)

RAM: 16 GB

Browser: Google Chrome / Microsoft Edge

4. Test Scenarios

Test Case	Description	Expected Result	Status
Dashboard Load Test	Open Tableau dashboard	Dashboard loads within a few seconds	Passed

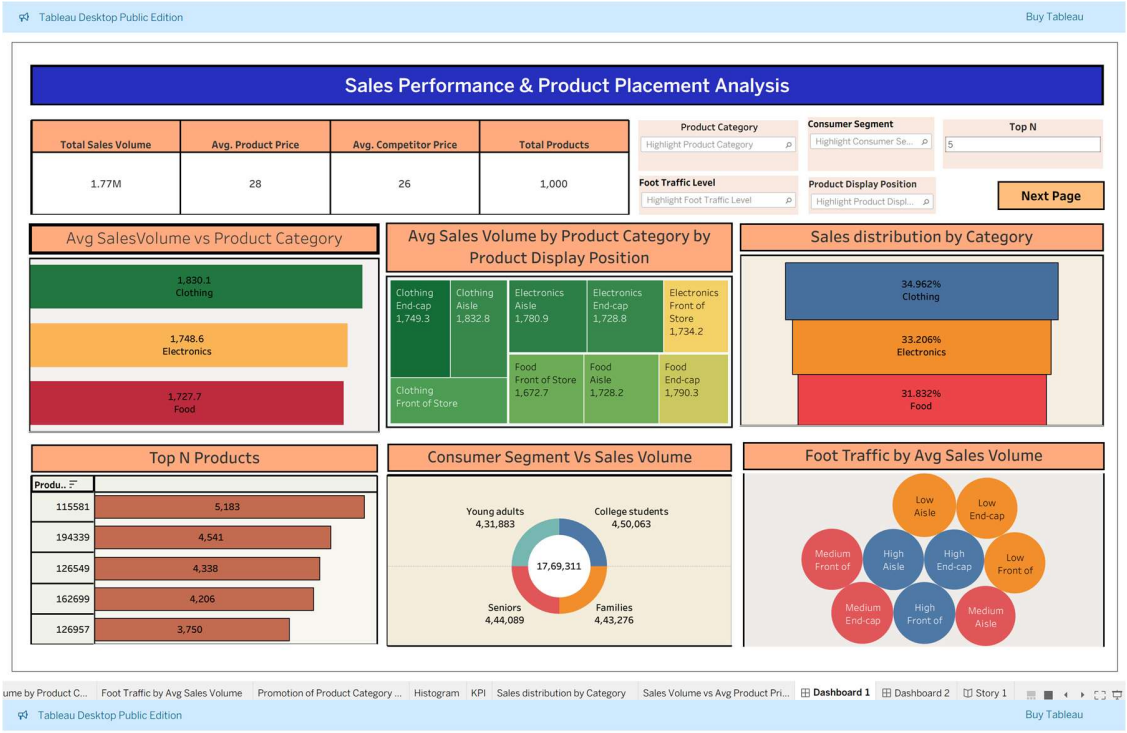
Filter Test	Apply Product Category filter	Charts update dynamically based on selection	Passed
Highlight Test	Select category highlight option	Selected category gets highlighted	Passed
Chart Accuracy Test	Verify labels and values	Displayed values match dataset	Passed
Interaction Test	Hover on charts for tooltips	Correct tooltip information appears	Passed

5. Testing Results

The Tableau dashboards performed efficiently during the testing phase. All charts and visualizations loaded properly without noticeable delay. Interactive filters such as Product Category and Consumer Segment responded correctly and updated the visualizations dynamically. The calculated metrics including Average Sales Volume and Price Difference were validated and found accurate.

6. Screenshots

Figure 1: Dashboard Overview



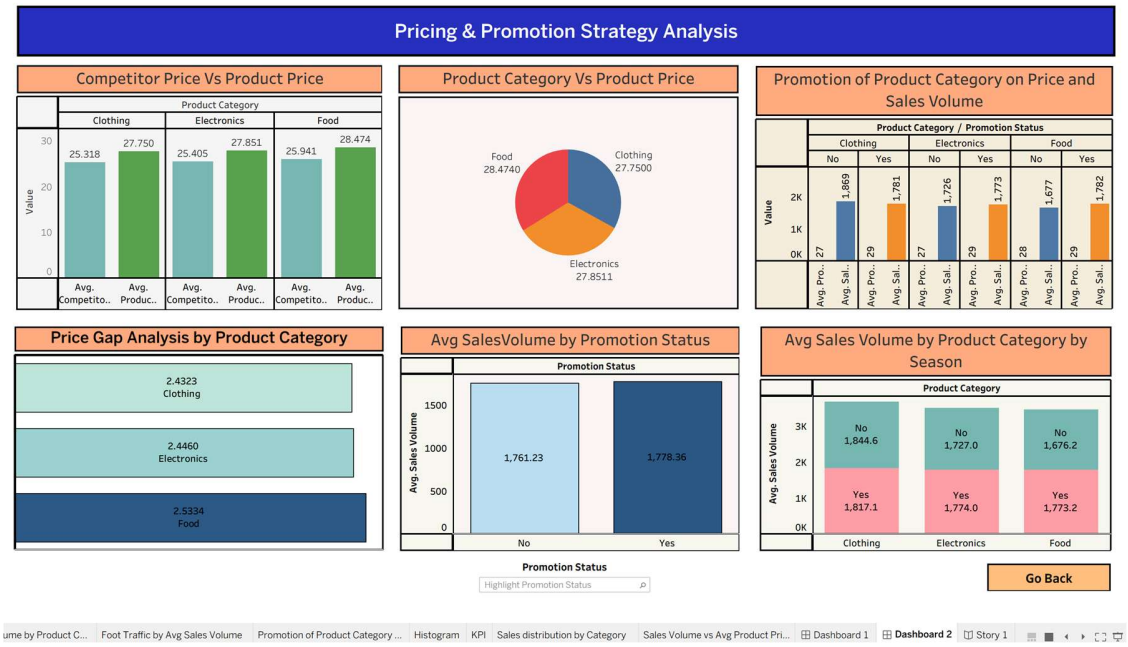


Figure 2: Filter Interaction Example

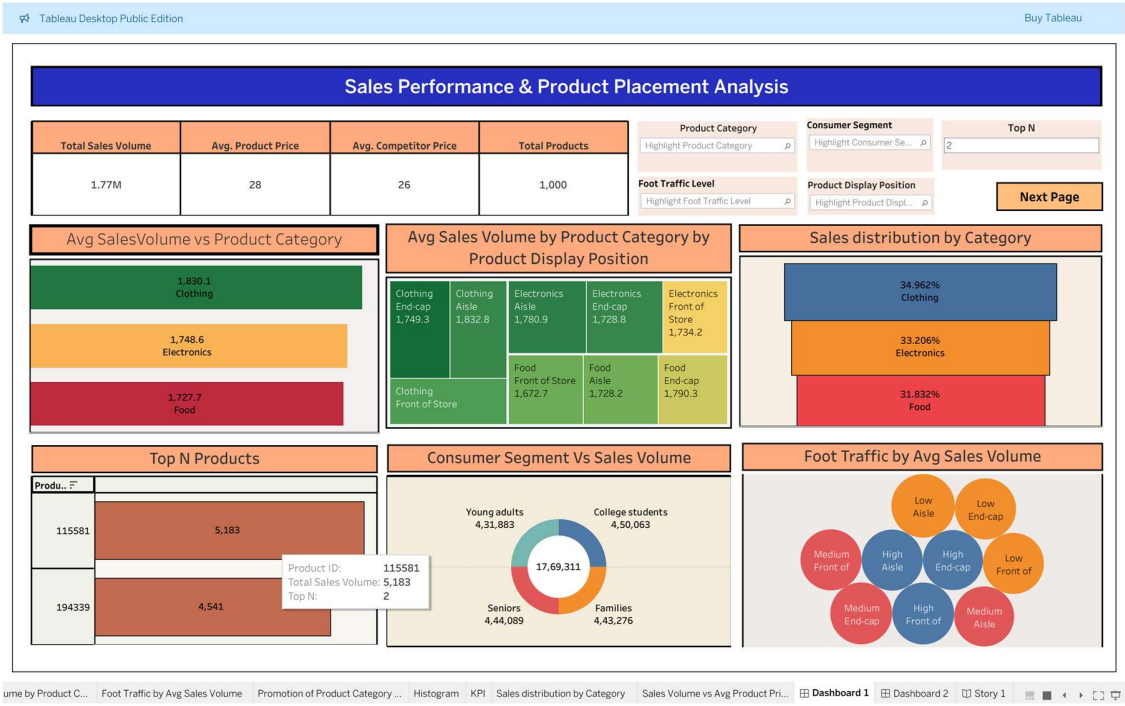
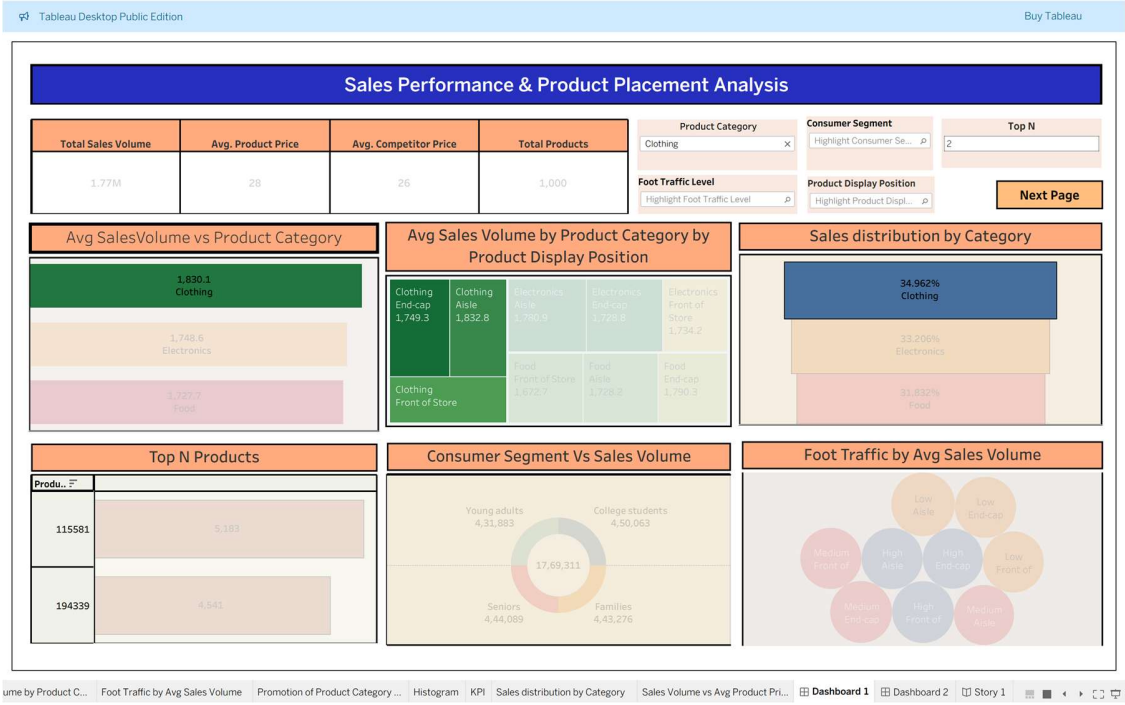


Figure 3: Category Highlight Example



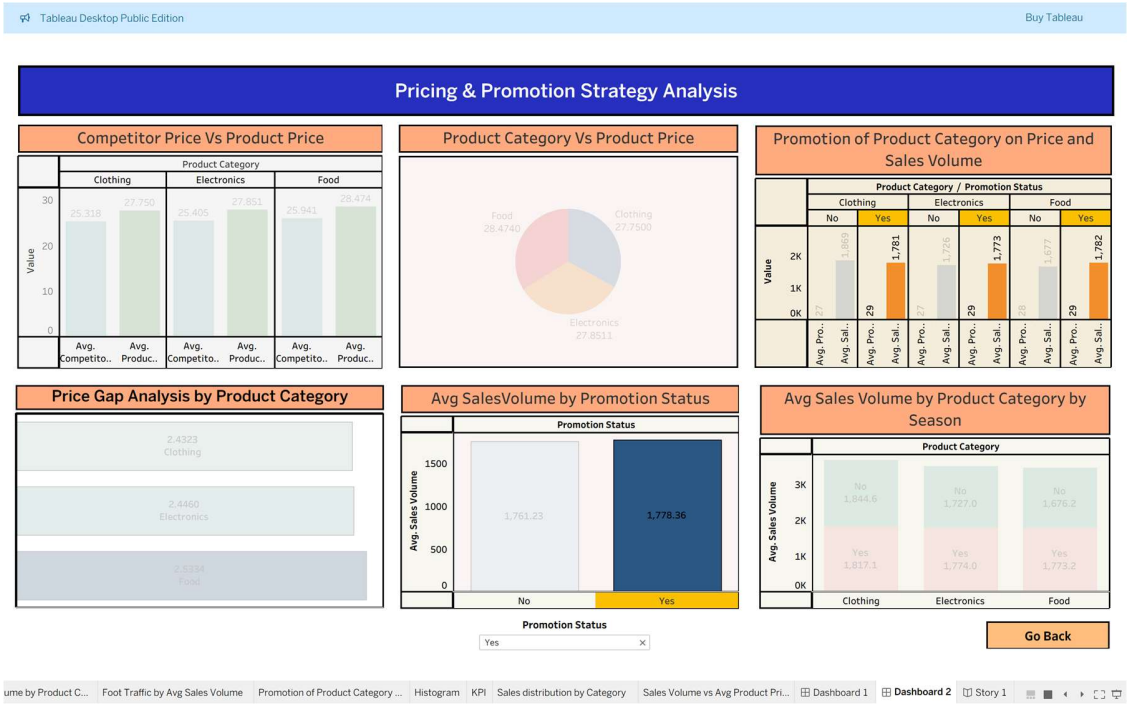


Figure 4: Chart Tooltip Interaction

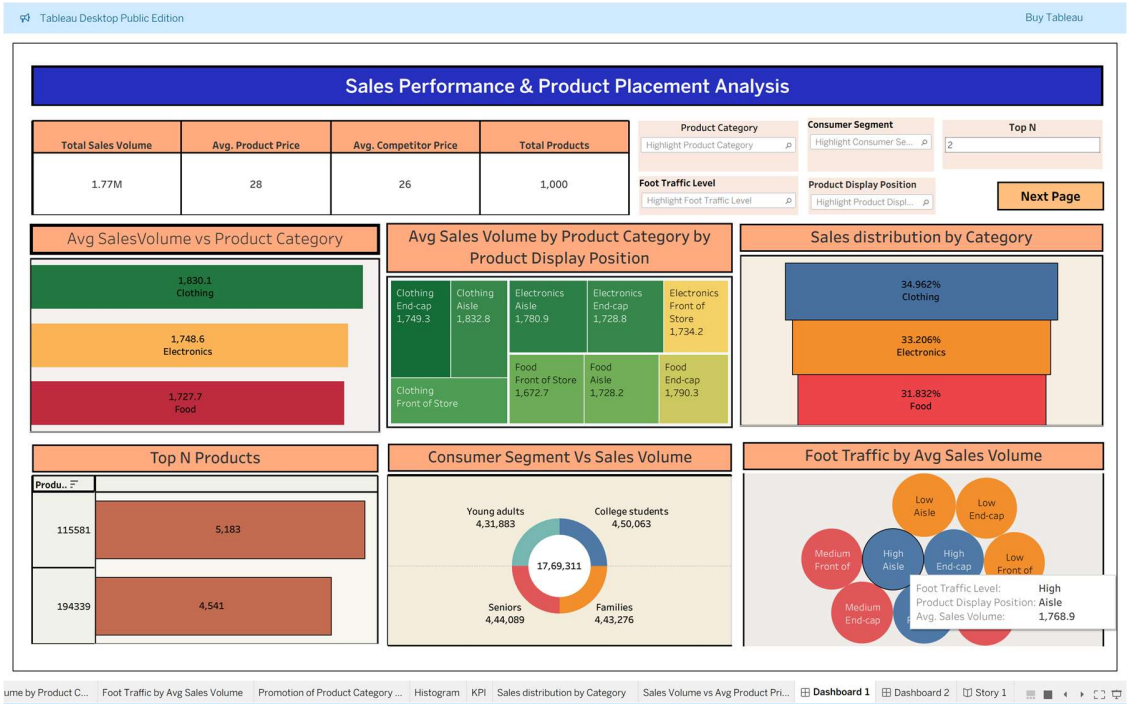


Tableau Desktop Public Edition
Buy Tableau
Sales Performance & Product Placement Analysis
Total Sales Volume
Avg. Product Price
Avg. Competitor Price
Total Products
1.77M
28
26
1,000
Product Category
Highlight Product Category
Consumer Segment
Highlight Consumer Se...
Top N
2
Foot Traffic Level
Highlight Foot Traffic Level
Product Display Position
Highlight Product Displ...
Next Page
Avg SalesVolume vs Product Category
1,830.1 Clothing
1,748.6 Electronics
1,727.7 Food
Avg Sales Volume by Product Category by Product Display Position
Clothing End-cap 1,749.3
Clothing Aisle 1,832.8
Electronics Aisle 1,780.9
Electronics End-cap 1,728.8
Electronics Front of Store 1,734.2
Clothing Front of Store
Food Front of Store 1,672.7
Food Aisle 1,728.2
Food End-cap 1,790.3
Sales distribution by Category
34.962% Clothing
33.206% Electronics
31.832% Food
Top N Products
Produ...
115581
5,183
194339
4,541
Consumer Segment Vs Sales Volume
Young adults 4,31,883
College students 4,50,063
17,69,311
Seniors 4,44,089
Families 4,43,276
Foot Traffic by Avg Sales Volume
Low Aisle
Low End-cap
Medium Front of
High Aisle
High End-cap
Low Front of
Medium End-cap
Foot Traffic Level: High
Product Display Position: Aisle
Avg. Sales Volume: 1,768.9
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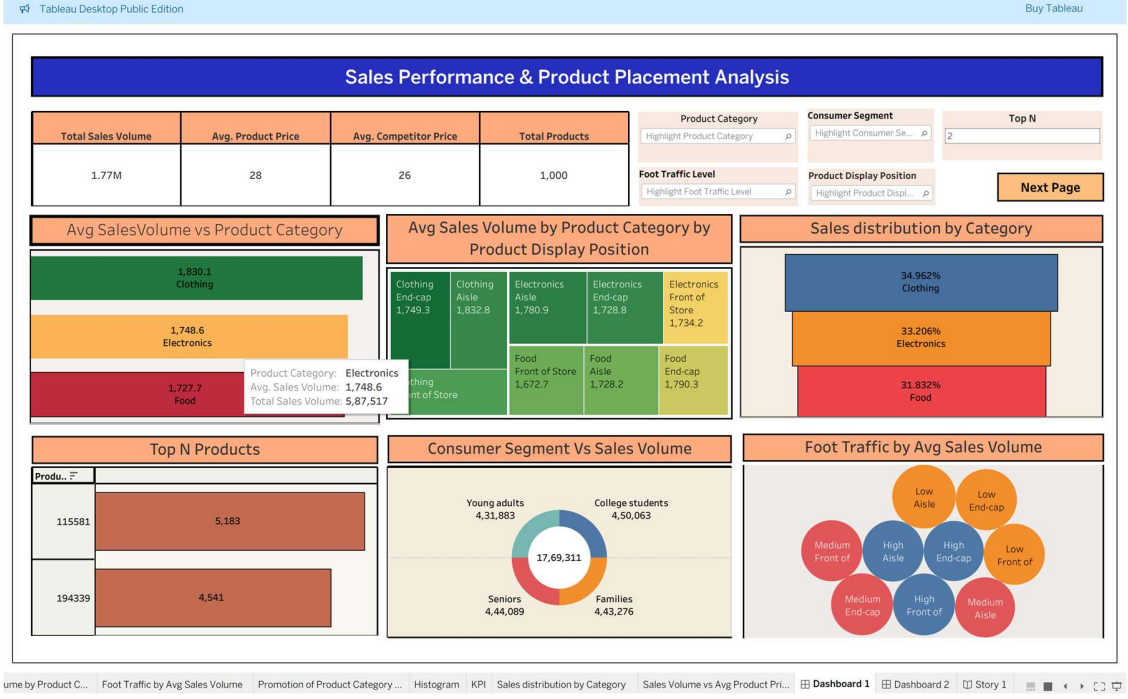
Foot Traffic by Avg Sales Volume
Promotion of Product Category ...
Histogram
KPI
Sales distribution by Category
Sales Volume vs Avg Product Pri...
Dashboard 1
Dashboard 2
Story 1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Avg. Sales Volume: 1,748.6
Total Sales Volume: 5,87,517
Avg Sales Volume by Product Category by Product Display Position
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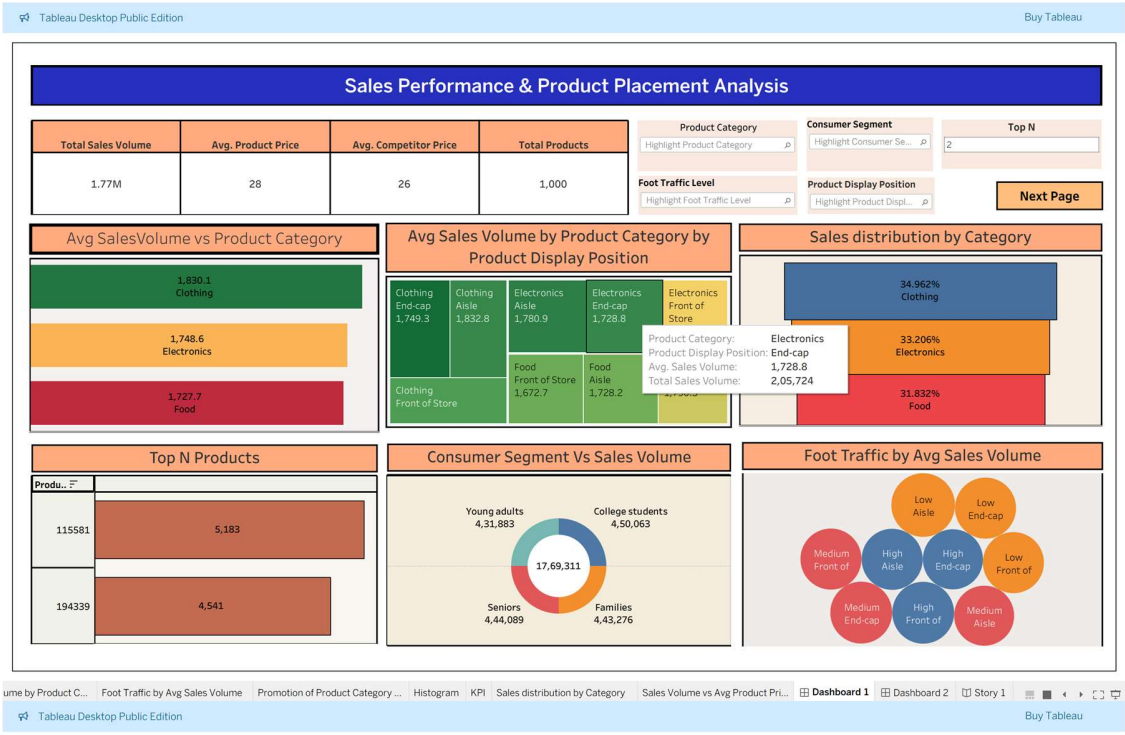

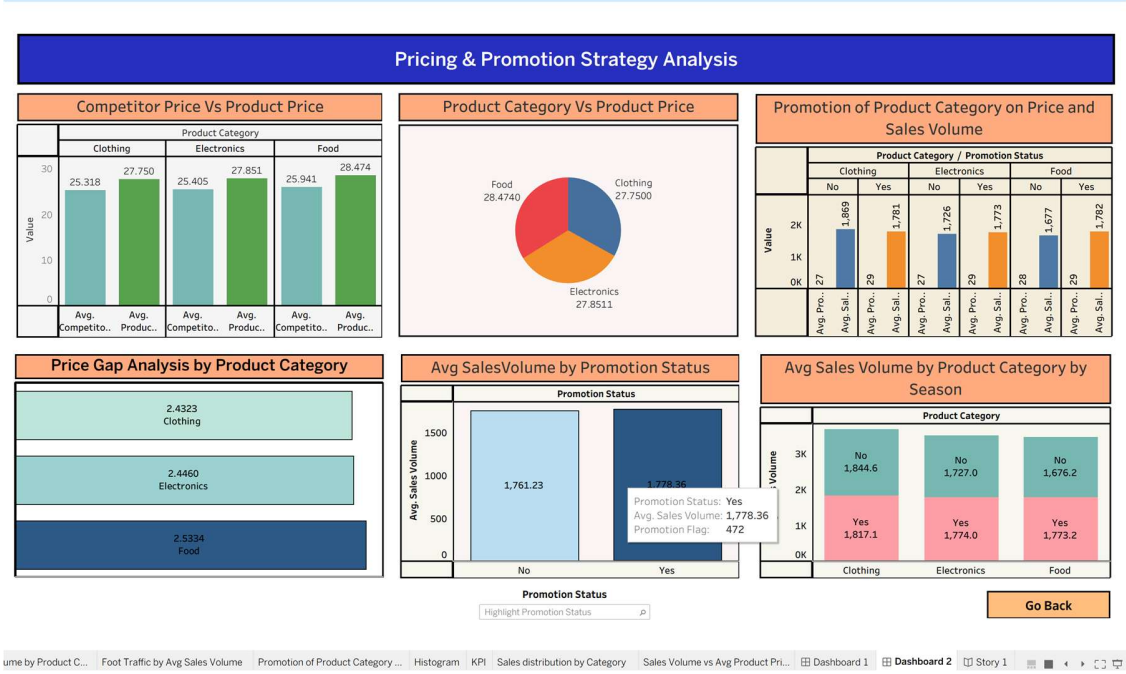

7. Conclusion

The performance testing confirmed that the Tableau dashboards are responsive, accurate, and user-friendly. All interactive features such as filters, highlights, and tooltips function correctly. The dashboards enable efficient analysis of product placement strategies and sales performance in retail environments.